Predicting star rating and sentiment from review text

Erdos Institute: Data Science Bootcamp Spring 2025

Group members: Duncan Clark, Andrew Silva, Brennan Register, Samantha Jarvis

Dataset construction

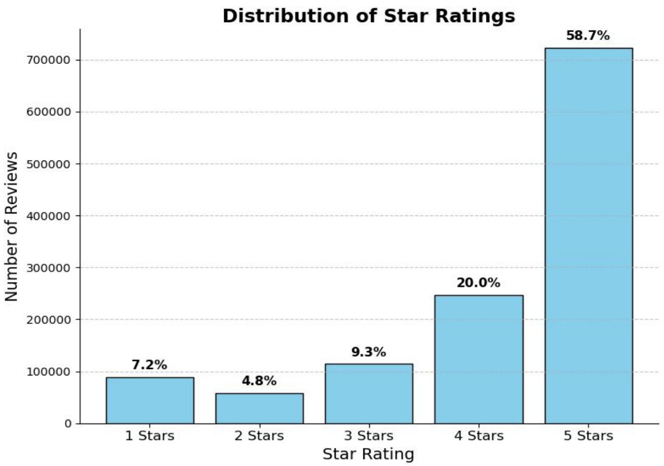
Google Local review data

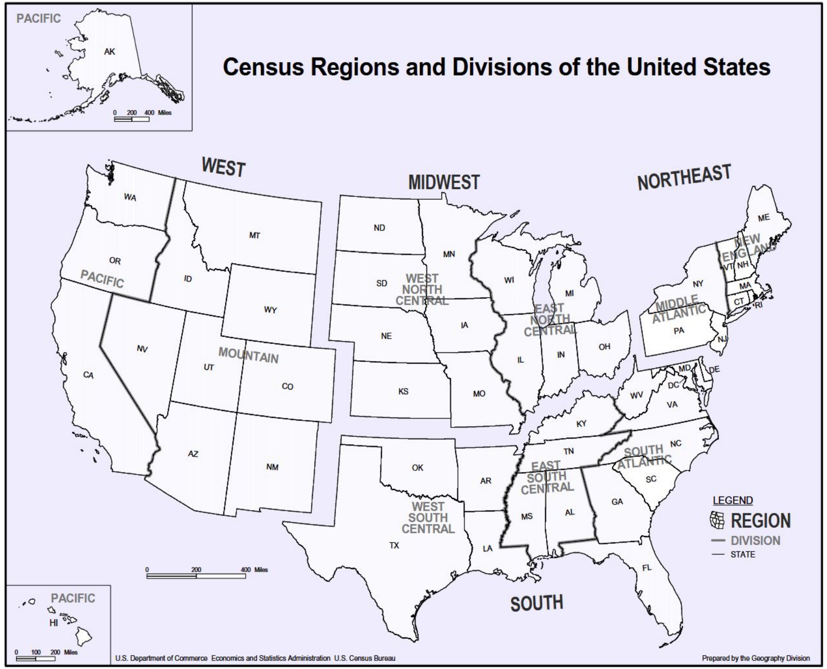
https://mcauleylab.ucsd.edu/public_datasets/gdrive/googlelocal/

Pre-collected dataset of Google reviews across the US

- Used metadata to identify reviews of restaurants from 10-core state files by matching location ID to review text
- Downsampled to 1% (~ 2 million total reviews) of total initial dataset, stratified by rating class and region (see right)
- Train/Val/Test split of 70%/15%/15% of original dataset

Mean rating: 4.183





Text processing pipeline

1. Preprocess review text

- Replace non-text characters (e.g., punctuation, emojis, “(Translated by Google)” tags, etc.)
- Make all words lowercase
- Slice sentence into list of tokens (words)

Examples

In: “This place will make your tummy smile 😀”
Tokens: [this, place, will, make, your, tummy, smile]
Vector: [21, 20, 59, 154, 82, 3886, 743, 0, … , 0]

In: “(Translated by Google) Ok pizza.\n\n”
Tokens: [ok, pizza]
Vector: [144, 63, 0, … ,0]

2. Create GloVe embedding layer

https://nlp.stanford.edu/projects/glove/

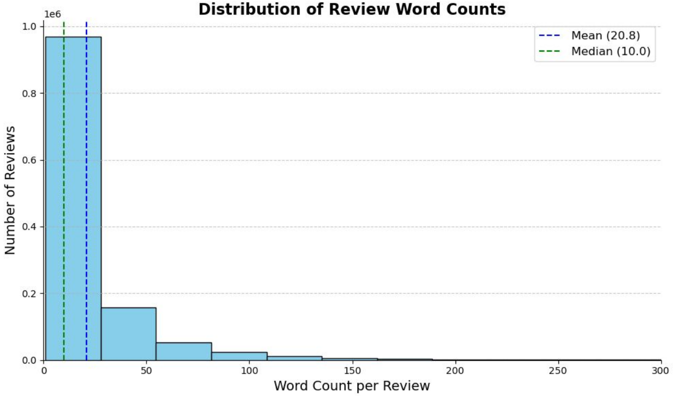
Pretrained model for dense embedding of words as vectors (100 dimensional space used). Semantic meaning is preserved by vector similarity (i.e., angle between vectors)

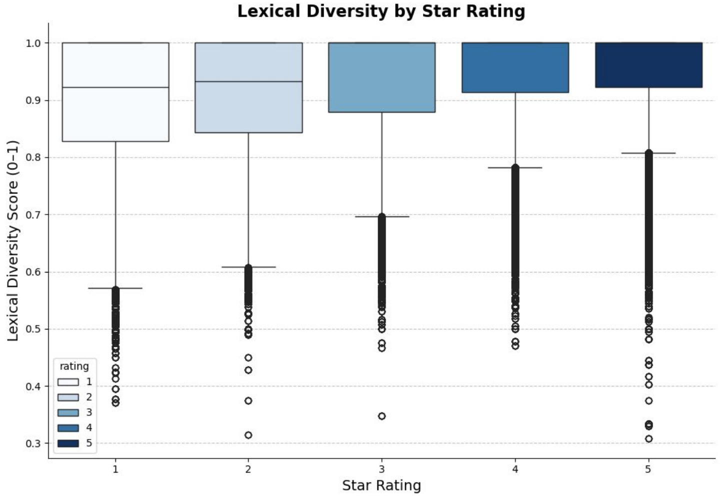
- GloVe vocabulary size = 400,000 words
- Used 20,000 most common words from the entire training set review text corpus to build embedding matrix
- Vector embeddings were frozen and not updated in training

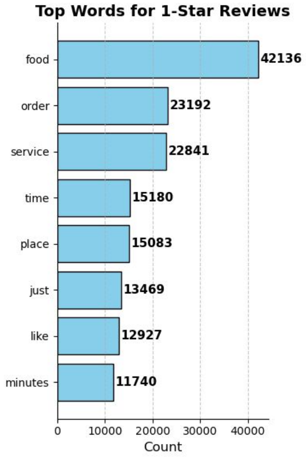
3. Encode text reviews as integer vectors

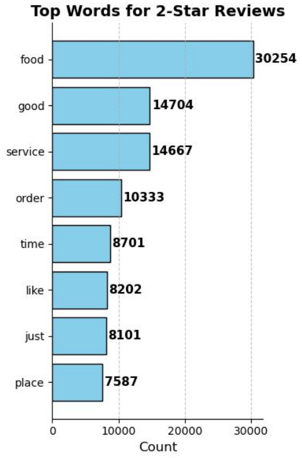
- List of tokens turned into vector whose entries correspond to words in the embedding matrix. Max length set to 128.
- Out of bounds tokens are skipped, 0s are used to pad sequences to max length.
- Reviews over 128 tokens in length are truncated.

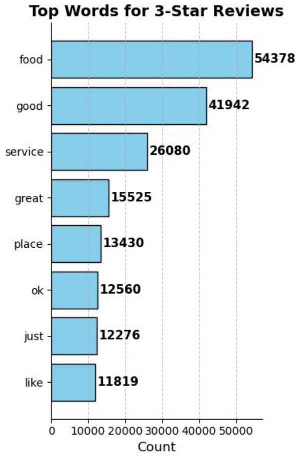
Descriptive summary of data

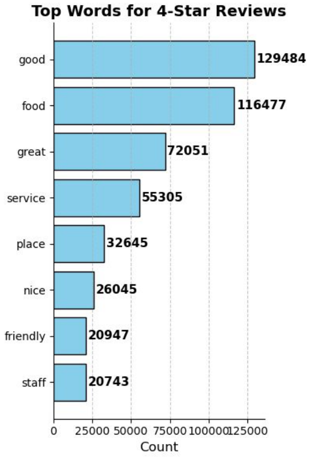


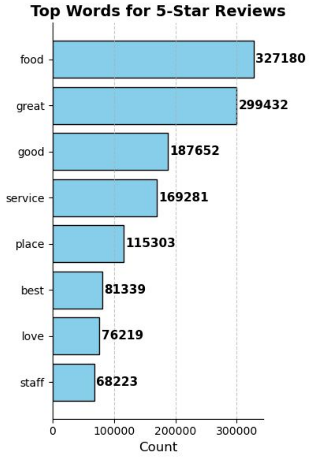












Models and training specifications

Model 1: Recurrent Neural Network (RNN) with Long Short Term Memory (LSTM)

- Embedding layer from GloVe (frozen for training)
- 2 hidden layers of dimension 196 with one linear classification layer (output is 3 or 5 dimensional)
- ~ 1.3 million trainable parameters
- Hidden layers are bidirectional LSTM with 30% dropout
- Saved model size ~10 MB

Model 2 (comparison): Cardiff-NLP RoBERTa

https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest

- RoBERTa-base model trained on ~124M tweets from January 2018 to December 2021, and fine-tuned for sentiment analysis with the TweetEval benchmark.
- ~ 124 million parameters
- Only classification layer is trained (~600,000 parameters)
- Saved model size: ~500 MB

Training specifications

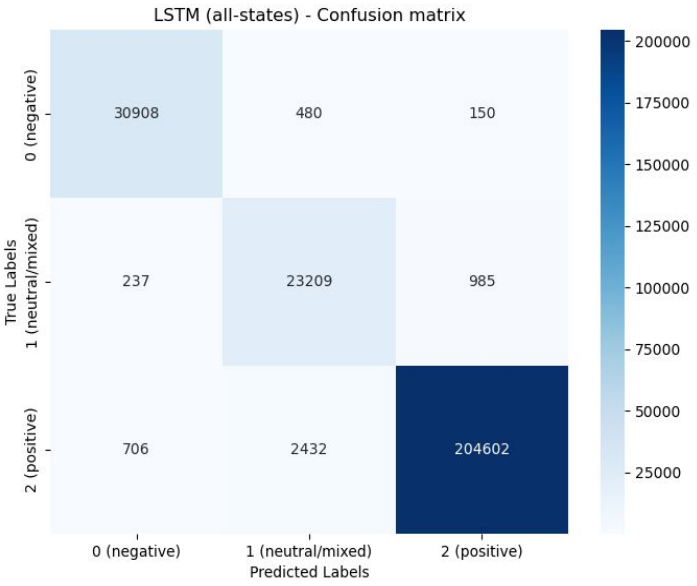
- Cross Entropy Loss used as classification criteria, class weights were applied due to large class imbalances in the dataset
  weights ratios approx. (59:20:9:7:5)
- Adams optimizer with Cosine Annealing learning rate scheduler
- Training halted early after 5 consecutive epochs of non-increasing F1 score
- Models trained locally on Nvidia RTX 3070 GPU with 8GM of VRAM

* For sentiment prediction, star ratings were grouped as:

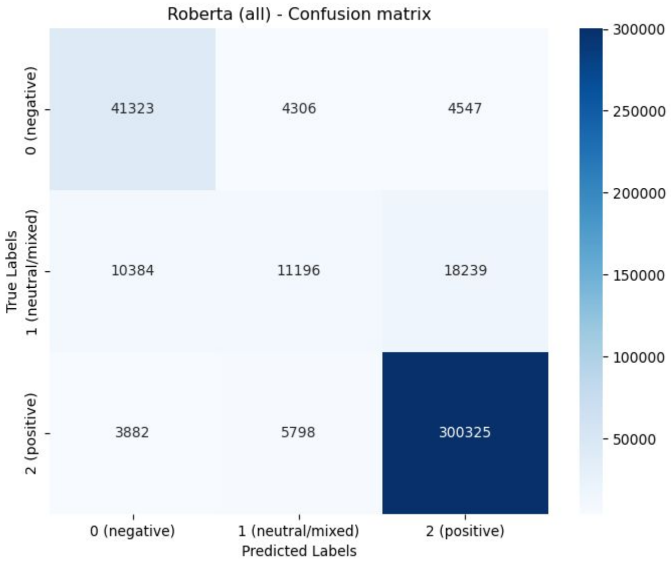
- 1, 2 = 0 “negative”
- 3 = 1 “neutral/mixed”
- 4, 5 = 2 “positive”

Results: sentiment prediction

LSTM Train time: 4 hrs | Prediction time: 30s
Accuracy: 98.11% | F1 score 0.98132



RoBERTa Train time: 9 hrs | Prediction time: 10 minutes
Accuracy: 89.06% | F1 score: 0.88166



*All models were trained locally on Nvidia RTX 3070 GPU with 8GB of VRAM

Results: predicting star ratings

LSTM used to predict star-rating based on review text

- Train duration: 9 hr 5 min | Prediction time: 30s
- **Accuracy**: **93.94% | F1 score: 0.94016**

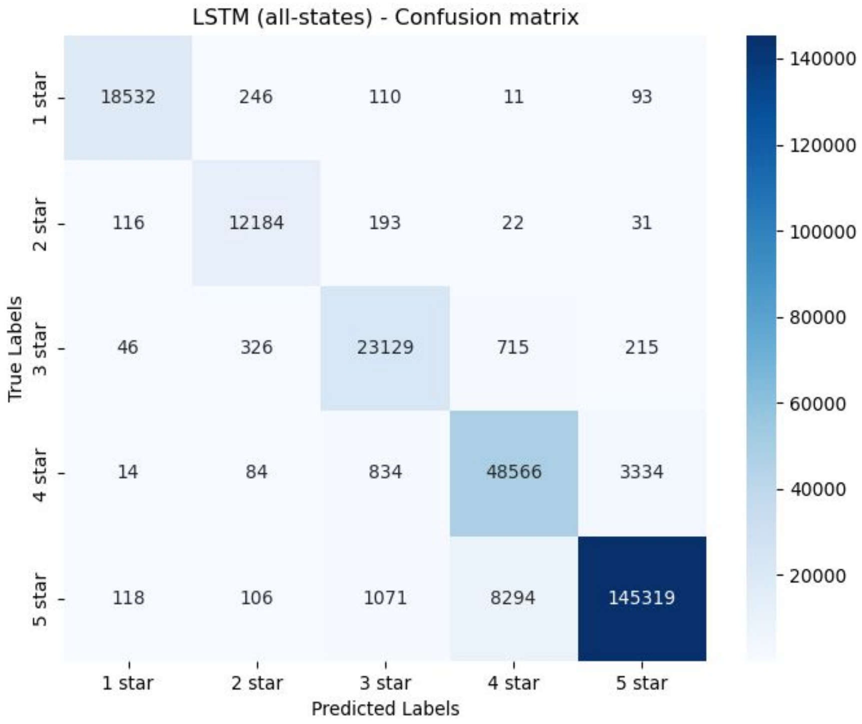
Example outputs:

Review: "'Absolutely disgusting! Only 5 pieces of Chicken and steak each and they were so dry i could barely swallow it…"
Actual rating: 1
Predicted rating: 1 star 99.9%

Review: "Very good indian food with a tinge of extra spice. Lunch is so cheap for a good quality food and enogh to make you sleepy!!"
Actual rating: 4 star
Predicted rating: 4 star 59.8% (5 star 40.1%)

Review: "Good service, boring decorations, tiny amount of fish on sandwiches, huge salads"
Actual rating: 3 star
Predicted rating 3 star 99.9%

Review: "It's pretty sad you go to Burger King and they only accept cash because they won't take your card"
Actual rating: 4 star
Predicted rating: 2 star 57.8% (1 star 41.8%)



Additional datasets

Vegetarian restaurants in the US Midwest

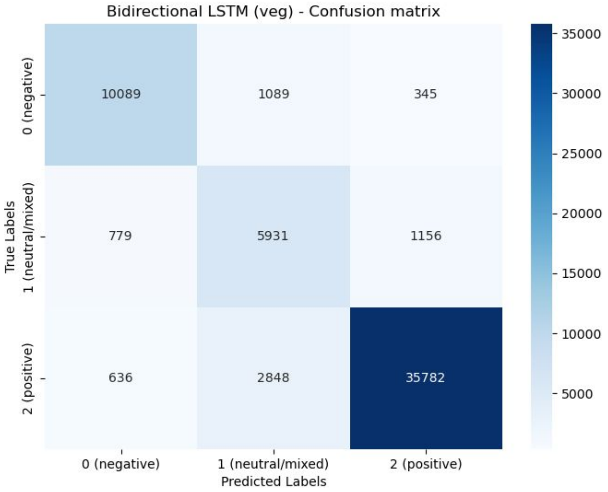
- train 480,135 samples
- val 96,027 samples
- test 58,655 samples

Mean rating: 3.802 (train)

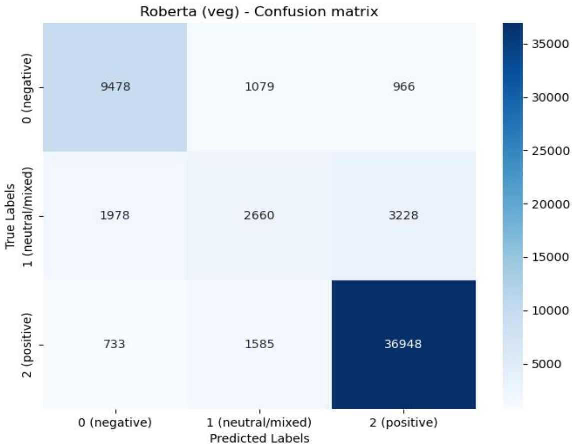
Rating distribution:

- 5 (45.7%)
- 4 (21.0%)
- 3 (13.5%)
- 2 (12.5%)
- 1 (7.22%)

LSTM: Train duration: 3.5 hrs
Accuracy: 88.32% | F1 score: 0.88769



RoBERTa: Train duration: 2.25 hrs
Accuracy: 83.69% | F1 score: 0.82592



Additional datasets

Thai restaurants in the US Midwest

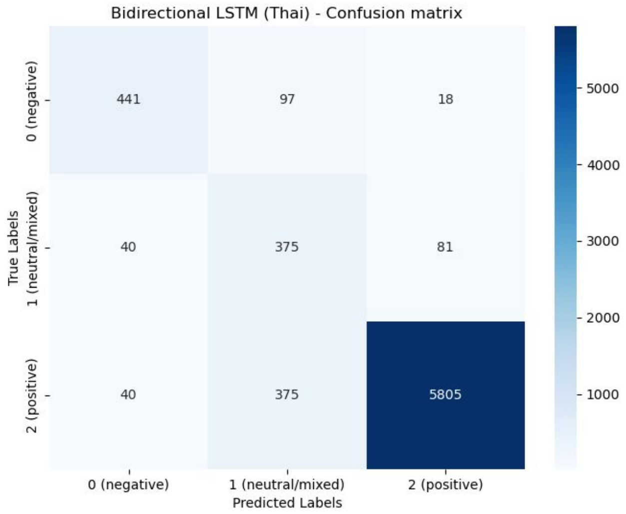
- train 66,494 samples
- val 16,577 samples
- test 7,272 samples

Mean rating: 4.398 (train)

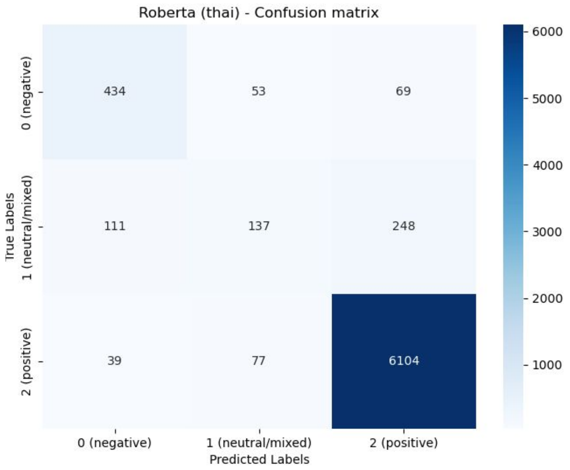
Rating distribution:

- 5 (65.8%)
- 4 (20.0%)
- 3 (6.69%)
- 2 (4.27%)
- 1 (3.33%)

LSTM: Train duration: 15 minutes
Accuracy: 91.05% | F1 score: >0.9



RoBERTa: Train duration: 27 minutes
Accuracy: 91.79% | F1 score: >0.9



Conclusions

Performance

- **LSTM** performed on-par with or better than **RoBERTa** with less training (roughly 50% duration)
- **LSTM** requires ~2% of the storage space of **RoBERTa**
- Prediction time of **LSTM** is ~5% that of **RoBERTa**

For the purposes of sentiment analysis of a specific type of establishment, the LSTM is more than sufficient

RoBERTa may have an edge in cases with spelling errors
Review: “Good food. Aweful service.”
LSTM prediction: Positive 80% (Mixed 19.6%, Negative .1%)
RoBERTa prediction: Mixed 50.8% (Positive 35.4%, Negative 13.7%)

Further considerations

- Could eek out better performance with additional text processing (e.g., emoji-to-text)
- Use RoBERTa or another transformer-based model ensemble with LSTM to improve performance
- Test robustness of LSTM by evaluating its predictive power for a different type of establishment from Google reviews (e.g., furniture stores)
- Use as further filtering on restaurant ratings to create recommendations based on review sentiment